

Predicting Gentrification Types from Housing Market Dynamics: A Machine Learning Classification Approach Using Census-Level Price Data in Spain

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Abstract

This paper develops a machine learning framework for predicting future gentrification processes at the census-section level across Spain. Using quarterly housing price data from the Idealista portal (2012–2022) covering 35,960 census sections, we operationalize gentrification into four distinct types based on the joint dynamics of rental and sale prices: renter pressure (Case A), speculative investment (Case B), active gentrification (Case C), and neighborhood degradation (Case D). We frame the problem as genuine forecasting: given observable market signals at time t , predict which gentrification type will manifest at $t+1$. After systematic comparison of eleven algorithms spanning gradient boosting, neural networks, SVMs, and recurrent models, an XGBoost pipeline with rolling-origin retraining and directional decomposition achieves the highest performance: 48% accuracy on the five-class task ($2.4\times$ the 20% chance baseline) and 0.77 AUC on binary displacement-risk detection. Under selective prediction, where the model abstains on uncertain cases, accuracy reaches 59% at 50% coverage and 65% at 30% coverage. Current gentrification state is the single most predictive feature, confirming strong temporal persistence, while external socio-economic data adds negligible predictive power beyond price dynamics alone. These results demonstrate that gentrification trajectories leave detectable signals in housing market data, offering a practical triage tool for targeted urban policy intervention.

Keywords: gentrification, machine learning, housing market, classification, XGBoost, census sections, Spain

1 Introduction

Residential areas are active forces that reproduce inequality (Park, 1916; Wacquant, 2005). In Spain, the housing sector has been a central economic pillar: between 1996 and 2007, prices appreciated by up to 232% in real terms (Arellano & Bentolila, 2009). The 2008 crisis triggered a collapse with unemployment reaching 26% by 2013 (Naredo, 2010), followed by recovery from 2014 onward (Alves & Urtasun, 2019). Against this backdrop, gentrification has emerged as a significant phenomenon, driven by neoliberal policies promoting private investment (Janoschka, Sequera, & Salinas, 2014; Sequera, 2013) and displacing lower-income residents (Slater, 2011).

This study addresses whether machine learning can predict the type of gentrification process that will emerge in a census section based on currently observable market signals.

We frame this as a multiclass classification problem with four theoretically grounded types, providing an early-warning system for urban policymakers.

2 Theoretical Framework

Gentrification, the transformation of a working-class area into one serving middle-class use (Sequera, 2013), is driven by the rent gap between past low investment and potential future returns (Smith, 2005). Hackworth and Smith (2001) showed how decentralization pressures local governments to promote gentrification for fiscal revenue, while Slater (2011) centered the human impact of displacement. Each census section constitutes its own housing market with distinct dynamics (Himmelberg, Mayer, & Sinai, 2005), motivating the granularity adopted here. The regional heterogeneity documented by Larraz-Iribas and Alfaro-Navarro (2008) underscores the importance of sub-national analysis.

3 Data and Operationalization

3.1 Data Source

The dataset consists of quarterly average listing prices (euros per square meter) for residential sale and rental properties collected by the Idealista portal from 2012 to 2022, organized at the census-section level (*seccion censal*), Spain’s smallest administrative unit. It contains approximately 1.46 million observations across 35,960 census sections, recording average unit prices and housing stock for both market segments. The census-section granularity allows detecting localized processes that aggregate data would obscure (Bruno Cueto, Mayor Fernández, & Suarez Pandiello, 2023; Himmelberg et al., 2005). Approximately 5% of sale-price and 28% of rental-price observations are missing, handled by exclusion.

3.2 Operationalization of Gentrification

Gentrification is operationalized through the joint dynamics of rental and sale price changes at the census-section level. The percentage change in both prices is computed quarter-over-quarter within each census section. Based on the direction of these changes, each observation is classified into one of four types:

- **Case A (Renter pressure):** Rent increases, sale prices stable or declining, signaling displacement pressure on tenants (Aja, 2007).
- **Case B (Speculative investment):** Sale prices increase, rents stable or declining, suggesting investor entry (Smith, 2005).
- **Case C (Active gentrification):** Both prices increase, indicating the consolidation phase (Hackworth & Smith, 2001).
- **Case D (Degradation):** Both prices decline, creating the rent gap that attracts future investment (Smith, 2005).

Conditions are evaluated hierarchically: C first (both positive), then D (both negative), then A and B.

4 Methodology

4.1 Prediction Framework

This study frames gentrification as a genuine forecasting task: given observable market features for a census section at time t , predict which gentrification type will manifest at $t+1$. Because the target is the *next* quarter’s case, all current-period features are legitimate inputs containing no information about future price changes. We also train binary classifiers for three policy-relevant targets: displacement risk (Cases A or C), active gentrification (Case C), and investment pressure (Cases B or C).

4.2 Feature Engineering

The base feature set consists of 50 variables organized into nine groups: current prices and their ratio; current case type (ordinal and one-hot indicators); lagged prices (1, 2, 4 quarters); lagged changes and their interaction; rolling means and standard deviations (3 and 6 quarters); momentum and acceleration; position relative to provincial mean; year-over-year changes; and housing stock levels, ratios, and temporal indicators.

Three additional feature groups capture temporal, spatial, and structural dynamics. *History features* (11 variables) encode each section’s expanding-window trajectory: cumulative case-type frequencies, flip rate, and rent/sale volatility, all lagged one period to prevent leakage. *Spatial features* (13 variables) capture neighborhood context at the district level: mean neighbor price changes, case-type distribution, price gaps, and municipality averages. *Mean-reversion features* (15 variables) exploit the anti-persistent structure of quarterly price changes (rent autocorrelation = -0.26 , sale = -0.21 , cross-correlation = 0.01) by encoding gap-to-rolling-mean, expanding z-scores, shock magnitude, sign streaks, and trend slopes.

4.3 Classification Models

We compare eleven algorithms across four families: gradient boosting (XGBoost (Chen & Guestrin, 2016), LightGBM, CatBoost, Random Forest, Voting/Stacking ensembles), neural networks (MLP), linear models (logistic regression, SVM), and LSTM networks (4- and 8-quarter windows). XGBoost hyperparameters were tuned via Bayesian optimization (Optuna, 50 trials) on a held-out 2018–2019 validation set.

4.4 Model Enhancements

Three techniques are applied to the best-performing XGBoost classifier. *Rolling-origin re-training* addresses concept drift by refitting the model for each test quarter using all observations with period $< t$, so the classifier adapts to evolving market dynamics (including the COVID-19 disruption) without any information leakage. *Regression meta-features* add out-of-time forecasts of next-quarter rent and sale price changes as four additional inputs: two continuous predictions and two sign indicators. These are produced by XGBoost regressors trained on expanding annual folds (train on years $< Y$, predict year Y), ensuring strict temporal separation. *Directional decomposition* replaces the monolithic five-class classifier with two three-class models that predict price direction (down, zero, up) for rent and sale separately; the five gentrification probabilities are recovered by

multiplying the constituent direction probabilities according to the case definitions (e.g., $P(A) = P(\text{rent up}) \times [P(\text{sale down}) + P(\text{sale zero})]$, $P(C) = P(\text{rent up}) \times P(\text{sale up})$).

4.5 External Data Integration

To test whether socioeconomic context improves prediction, we integrate seven INE datasets: household income (ADRH), Census 2011 demographics, unemployment (EPA), mortgages, tourism, CPI, and foreign population.

4.6 Evaluation Strategy

The dataset is split temporally: pre-2020 observations ($\sim 798,000$ rows) for training and 2020+ ($\sim 199,000$ rows) for testing, preventing information leakage and testing generalization across the COVID-19 disruption. A separate 2018–2019 validation set (trained on pre-2018 data) is used for all hyperparameter and design decisions, including ensemble weights, composition blend coefficients, and feature selection thresholds. Evaluation uses accuracy, weighted F1, and AUC-ROC (binary tasks). For selective prediction, observations are ranked by the model’s maximum predicted probability and only the top fraction is retained, producing an accuracy–coverage tradeoff curve.

5 Results

5.1 Case Distribution

Among the five classified types, Case B (speculative investment) is the most frequent (27.6%), followed by A (19.7%), D (19.2%), C (18.8%), and None (14.8%). The temporal evolution (Figure 1) traces a structural narrative: Case D dominates in 2012–2013 (post-crisis), Case B expands by 2014–2016, and Case C grows steadily from 2017 as speculation matures into full gentrification.

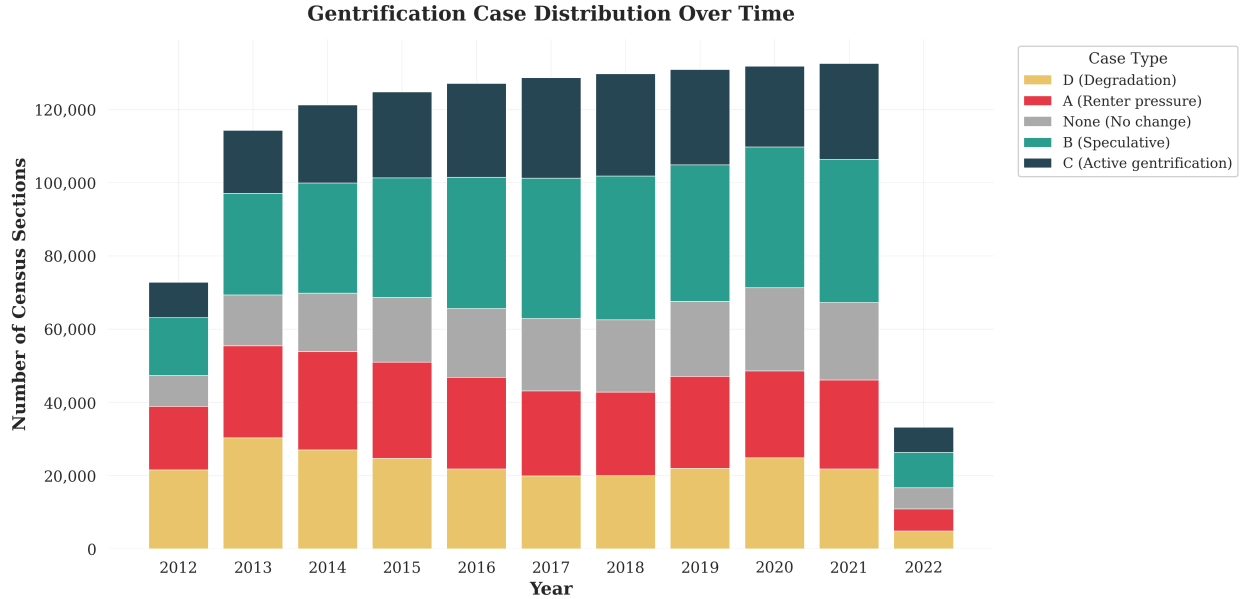


Figure 1: Distribution of gentrification cases by year. The shift from degradation (D) in 2012–2013 to speculative investment (B) and active gentrification (C) in 2017–2021 traces the post-crisis recovery.

5.2 Algorithm Comparison

Table 1 presents the systematic comparison across all eleven algorithms on the five-class task using 35 price-based features (excluding current case type to enable fair comparison across model families). All gradient-boosting methods converge to approximately 38% accuracy, with XGBoost and the Voting Ensemble tied at 38.2%. Neural networks (MLP) reach 36.7%, linear models 34.5–34.8%, and LSTM recurrent networks only 30.0%, substantially below the tree-based methods.

Table 1: Five-class accuracy across algorithm families (35 features, no case type). All evaluated on the same temporal test set ($\sim 199,000$ obs). Chance baseline is 20%.

Family	Algorithm	Accuracy	AUC
Gradient boosting	XGBoost	0.381	0.736
	LightGBM	0.381	0.736
	CatBoost	0.381	0.735
	Random Forest	0.379	0.730
	Voting Ensemble	0.382	–
Neural network	MLP (256-128-64)	0.367	0.733
	MLP (128-64)	0.356	0.734
Linear	Linear SVM	0.348	–
	Logistic Regression	0.345	0.716
Sequence	LSTM (seq=4)	0.299	0.613
	LSTM (seq=8)	0.300	0.609

When current case type is added as a feature (50 features total), XGBoost with Bayesian-

optimized hyperparameters (Optuna, 50 trials on a held-out 2018–2019 validation set) reaches 42.9% accuracy (Table 2), a 4-percentage-point improvement confirming the strong temporal persistence of gentrification dynamics.

Table 2: XGBoost performance with progressive enrichment (2020+ test set). Chance baseline is 20%.

Model variant	Features	Accuracy	Weighted F1
Baseline (no case type)	35	0.381	0.374
+ current case type	50	0.427	0.424
+ history & spatial features	74	0.442	0.437
+ mean-reversion geometry	89	0.449	0.446
+ rolling-origin retraining	89	0.472	0.470
+ regression meta-features	99	0.477	0.476
+ directional decomposition	99	0.479	0.478

Each enrichment step yields cumulative gains. History features add 0.5pp, spatial features 1.0pp, and mean-reversion features 0.7pp beyond the case-type baseline. Rolling-origin retraining contributes the largest single gain (+2.3pp) by adapting to distributional shift across the COVID-19 disruption. Regression meta-features add 0.5pp, and directional decomposition adds 0.2pp by aligning the classification boundary with the gentrification type definitions. Per-class precision and recall are balanced (0.43–0.55 range); Case B achieves the highest recall (0.55).

5.3 Binary Early-Warning Performance

Table 3 reports binary classifier performance for three policy-relevant aggregations. The displacement-risk detector (Cases A or C) achieves AUC 0.77, correctly flagging 70% of displacement events while maintaining 75% precision on non-displacement predictions.

Table 3: Binary early-warning performance (2020+ test set).

Target	Accuracy	AUC	Positive recall
Displacement risk (A C vs rest)	0.696	0.770	0.70
Active gentrification (C vs rest)	0.662	0.731	0.67
Investment pressure (B C vs rest)	0.628	0.686	0.67

5.4 Feature Importance and External Data

The feature importance ranking (Figure 2) reveals that current gentrification case type dominates all other features, confirming strong temporal persistence. Among purely market-based features, rental stock, the rent-to-sale ratio, and current price changes are the strongest predictors, consistent with rental markets responding faster to neighborhood change (Slater, 2011).

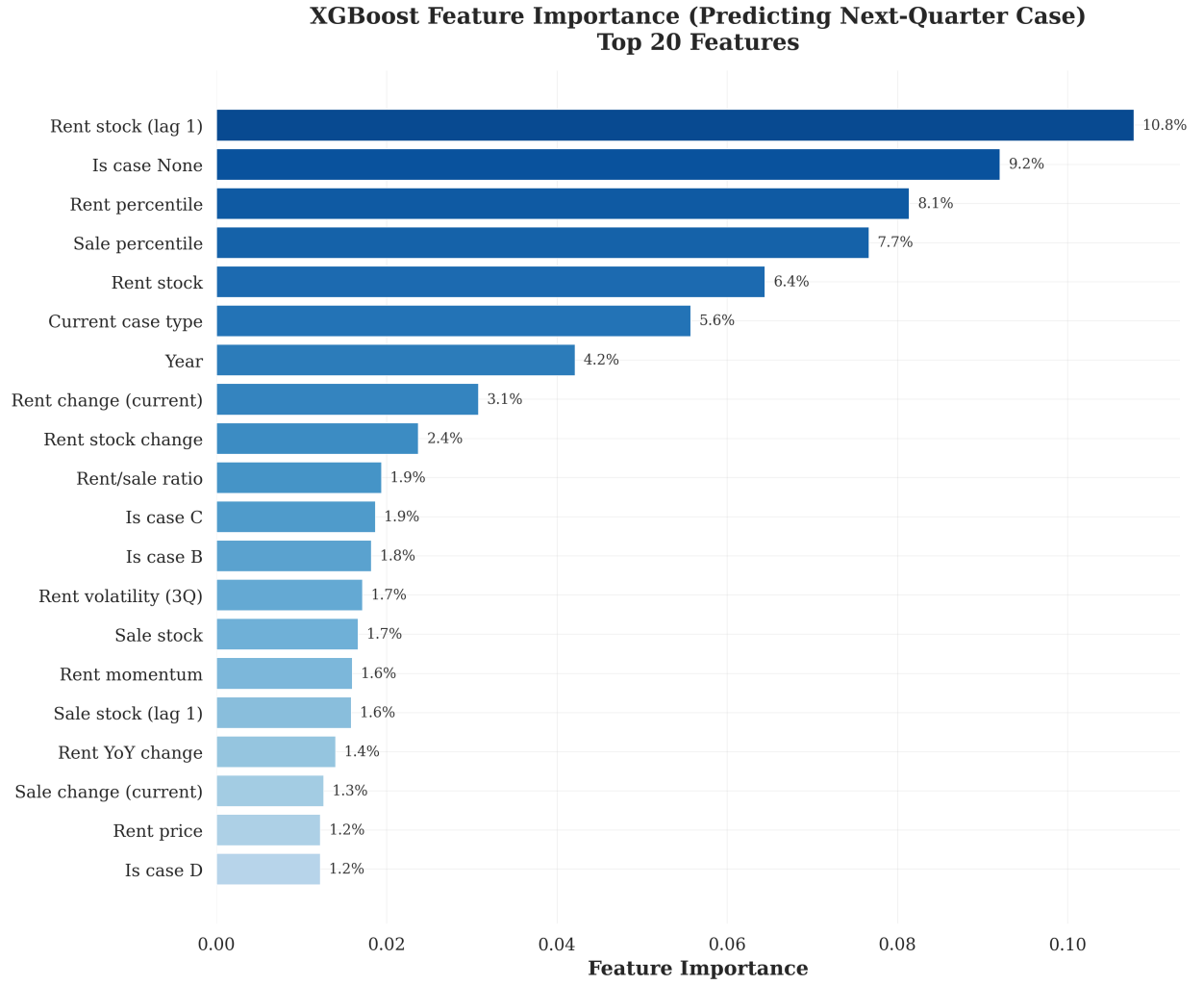


Figure 2: Feature importance from the XGBoost classifier. Current case type dominates, confirming strong persistence in gentrification dynamics.

Integration of all seven external datasets adds only +0.24pp over the price-only baseline, with no external feature among the top 10 most important. Price dynamics already encode the socioeconomic information relevant for one-quarter-ahead prediction.

5.5 Geographic Patterns

Despite 48% per-observation accuracy, the model correctly recovers the dominant gentrification type in 47 of 50 provinces (94%, excluding the Canary Islands), with province-level Pearson $r = 0.92$ (Figure 3). Prediction errors are distributed across classes rather than systematically biased. Madrid, Barcelona, and the Balearic Islands show the highest Case C intensity, consistent with their roles as primary destinations for real estate investment.

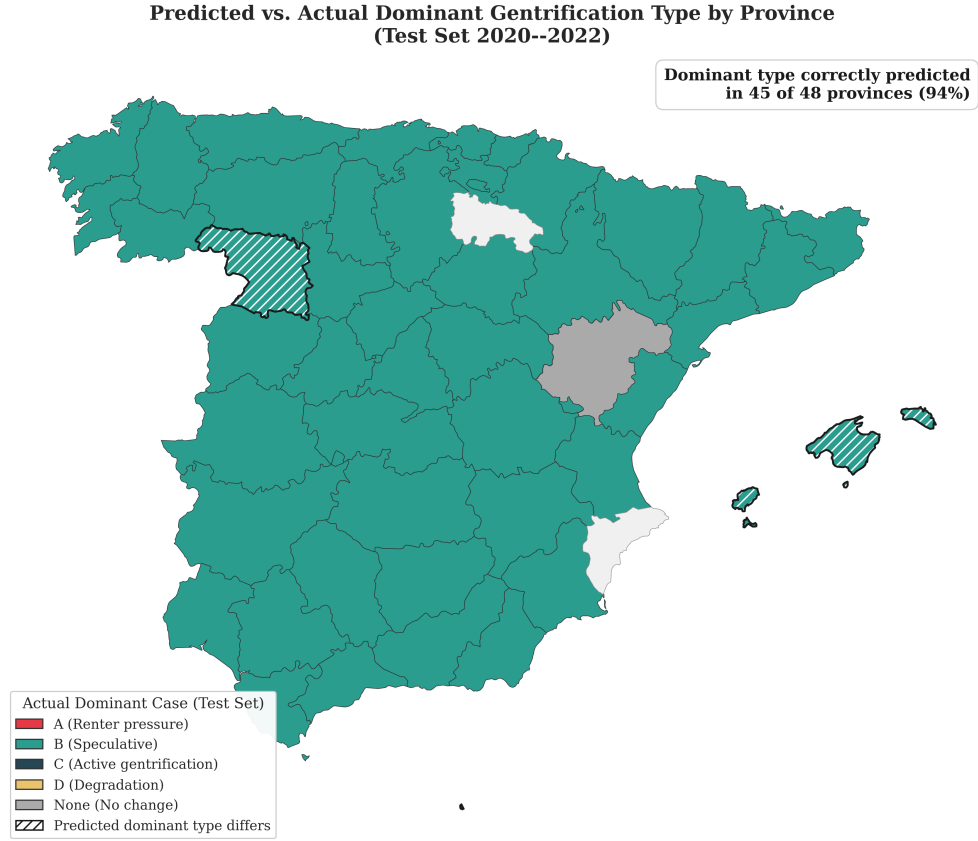


Figure 3: Predicted versus actual dominant gentrification type by province (2020–2022). Hatching marks disagreement.

5.6 Prediction Horizon

Extending the forecasting horizon beyond one quarter reveals rapid accuracy decay: from 38.2% at $t+1$ to 30.0% at $t+2$ (two quarters) and 28.2% at $t+4$ (one year). By four quarters ahead, accuracy approaches the 20% chance baseline, indicating that the predictive signal in quarterly price changes is inherently short-range.

5.7 Selective Prediction

In practical deployment, a policy tool need not predict every census section; it can abstain on uncertain cases and focus resources on high-confidence predictions. Table 4 shows the accuracy–coverage tradeoff when ranking predictions by the model’s maximum predicted probability and retaining only the top fraction.

Table 4: Selective prediction: accuracy at different coverage levels (final model, 2020+ test set).

Coverage	Accuracy	Sections predicted
100%	0.479	199,000
80%	0.520	159,000
60%	0.562	119,000
50%	0.586	99,500
40%	0.613	79,600
30%	0.647	59,700

At 50% coverage, accuracy reaches 58.6%, well above the 50% threshold that represents correctly predicting two independent binary signs simultaneously ($r = 0.01$). At 30% coverage, accuracy rises to 64.7%. This result reframes the practical value of the model: rather than an imperfect census-wide predictor, it becomes a reliable triage tool that identifies the roughly 100,000 sections per quarter where its predictions are trustworthy.

An annual reformulation (predicting year $Y+1$ from annual summaries of year Y) achieves 42.4% at full coverage, confirming the baseline ceiling persists across resolutions.

5.8 Regional Distribution of Gentrification Types

Beyond aggregate accuracy, the model’s geographic structure reveals where each gentrification type concentrates. Figure 4 maps the province-level share of each case type across the full 2012–2022 period. The four panels reveal distinct spatial footprints that align with Spain’s economic geography.

Case A (renter pressure) concentrates along the Mediterranean corridor and in Spain’s two largest metropolitan areas: Cantabria (23.5%), Madrid (23.2%), Barcelona (21.9%), Zaragoza (21.6%), and Valencia (21.4%). These are provinces with high rental demand and large tenant populations, where rising rents without corresponding sale price appreciation signal displacement pressure on existing residents. Rural provinces such as Teruel (12.0%) and Cuenca (12.4%) show substantially lower rates.

Case B (speculative investment) is the most geographically widespread type, dominating 39 of 52 provinces as the most frequent case. It is particularly intense in the Basque Country (Gipuzkoa 31.0%, Araba 30.8%), Murcia (30.9%), and Andalusian provinces like Jaen (30.6%). The Basque pattern reflects high property values creating conditions for sale-driven speculation, while in southern and southeastern Spain it reflects post-crisis recovery dynamics where sale prices rebounded while rental markets remained stagnant.

Case C (active gentrification) shows the sharpest geographic concentration: the Balearic Islands (25.1%), Madrid (23.4%), and Malaga (23.1%) lead by wide margins. These three provinces share exposure to international real estate investment and tourism-driven demand that pushes both prices upward simultaneously. Barcelona (21.4%) and Valencia (21.0%) complete the top five. At the other extreme, rural provinces such as Teruel (9.2%), Cuenca (10.4%), and Caceres (11.7%) rarely experience synchronized price increases, consistent with weak demand in depopulating areas.

Case D (degradation) follows a spatial pattern that partially mirrors Case C: Madrid (22.8%), Cantabria (22.3%), Barcelona (21.7%), and the Balearic Islands (21.1%) also show the highest D rates. This co-occurrence is not contradictory; it reflects the cyclical nature of

urban housing markets in dynamic provinces, where periods of simultaneous decline alternate with periods of synchronized growth. Provinces with low market volatility (Teruel, Cuenca) show low rates of both C and D.

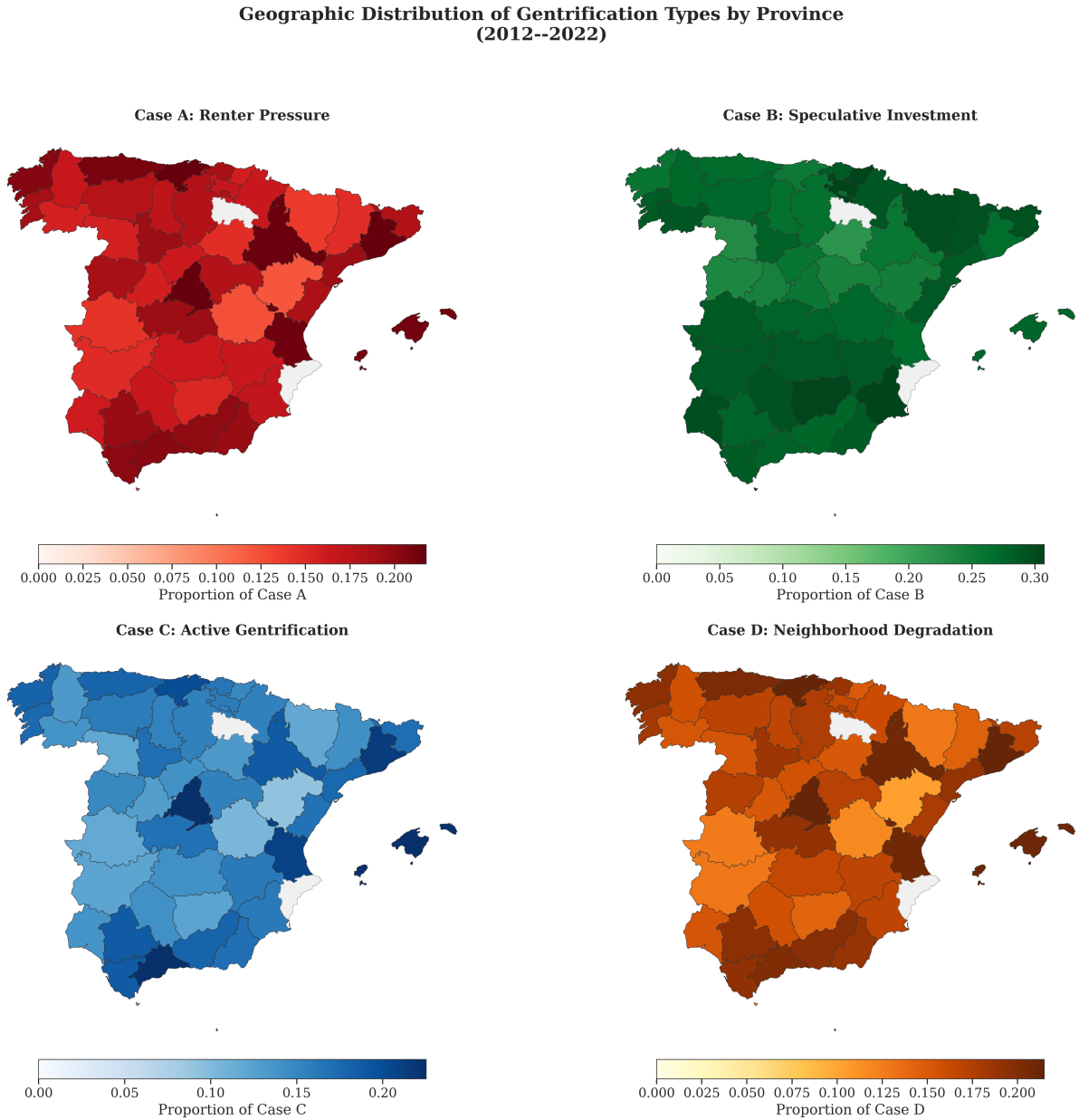


Figure 4: Province-level share of each gentrification type (2012–2022). Cases A and C concentrate in major metropolitan and coastal tourism areas; Case B is geographically ubiquitous; rural interior provinces show lower intensity across all active types.

Aggregating by region sharpens the contrast. The Mediterranean coast shows the highest combined A+C rate (41.1%), indicating that rental and sale markets move together in tourist-exposed areas, while the “None” rate is only 10.9%, reflecting persistent market dynamism. Northern Spain and Andalusia, by contrast, show elevated Case B rates (28.4–28.5%) and higher “None” shares (17–18%), suggesting markets where speculative sale activity proceeds without corresponding rental pressure. The interior provinces occupy a middle ground.

The temporal dimension adds a structural narrative. In 2012–2015, Case D dominated

11 provinces as the post-crisis collapse drove simultaneous price declines. By 2018–2021, no province retained D as its dominant type; Case B had expanded to 39 provinces, reflecting the nationwide recovery in sale prices. This transition from degradation to speculation follows the rent-gap cycle theorized by Smith (2005): crisis creates devaluation, which widens the gap between current use and potential returns, attracting the investment that initiates the next gentrification wave. The geographic selectivity of Case C, concentrated in just a handful of high-demand provinces, suggests that the transition from speculation (B) to full gentrification (C) requires external demand drivers (tourism, international capital, migration pressure) beyond simple price recovery.

5.9 District-Level Analysis: Madrid and Barcelona

Province-level analysis reveals the broad geography; district-level analysis within Spain’s two largest cities exposes the intra-urban structure. Madrid (21 districts, 2,408 census sections) and Barcelona (10 districts, 1,061 sections) together account for over 138,000 classified observations.

Madrid exhibits a clear center–periphery gradient (Figure 5). Central and upper-income districts are dominated by Case C (active gentrification): Centro (28.4% Case C), Salamanca (27.7%), Moncloa-Aravaca (26.9%), Arganzuela (26.8%), and Chamartín (26.7%). In these areas, both rental and sale prices increase simultaneously, reflecting sustained demand from higher-income residents and investors. Peripheral working-class districts are instead dominated by Case B (speculative investment): Usera (28.4% Case B), Villaverde (26.8%), Puente de Vallecas (26.6%), and Moratalaz (25.8%). Here, sale prices rise while rents remain stagnant, consistent with investor-driven property acquisition that has not yet translated into rental market pressure. Two districts show distinct patterns: Tetuán and Vicálvaro are dominated by Case A (renter pressure), while Fuencarral-El Pardo and Carabanchel are dominated by Case D (degradation).

Dominant Gentrification Type: Madrid Districts (2012--2022)



Figure 5: Dominant gentrification type by district in Madrid (2012–2022). Active gentrification (Case C) concentrates in central districts; speculative investment (Case B) dominates the periphery.

Barcelona shows a similar but more compressed pattern (Figure 6). The historic center leads in Case C: Ciutat Vella (26.4%), l'Eixample (26.2%), and les Corts (26.2%). Peripheral districts with lower-income populations are dominated by Case B: Nou Barris (25.7% Case B), Sant Andreu (25.4%), and Sant Martí (24.5%). An unexpected finding is that several western and hillside districts (les Corts, Sarrià-Sant Gervasi, Gràcia, Horta-Guinardó) are dominated by Case D, reflecting the high market volatility in Barcelona's wealthier neighborhoods, where price corrections alternate with growth phases. Sants-Montjuïc, the city's southernmost district with its mixed industrial and residential character, is the only district dominated by Case A.

Dominant Gentrification Type: Barcelona Districts (2012--2022)

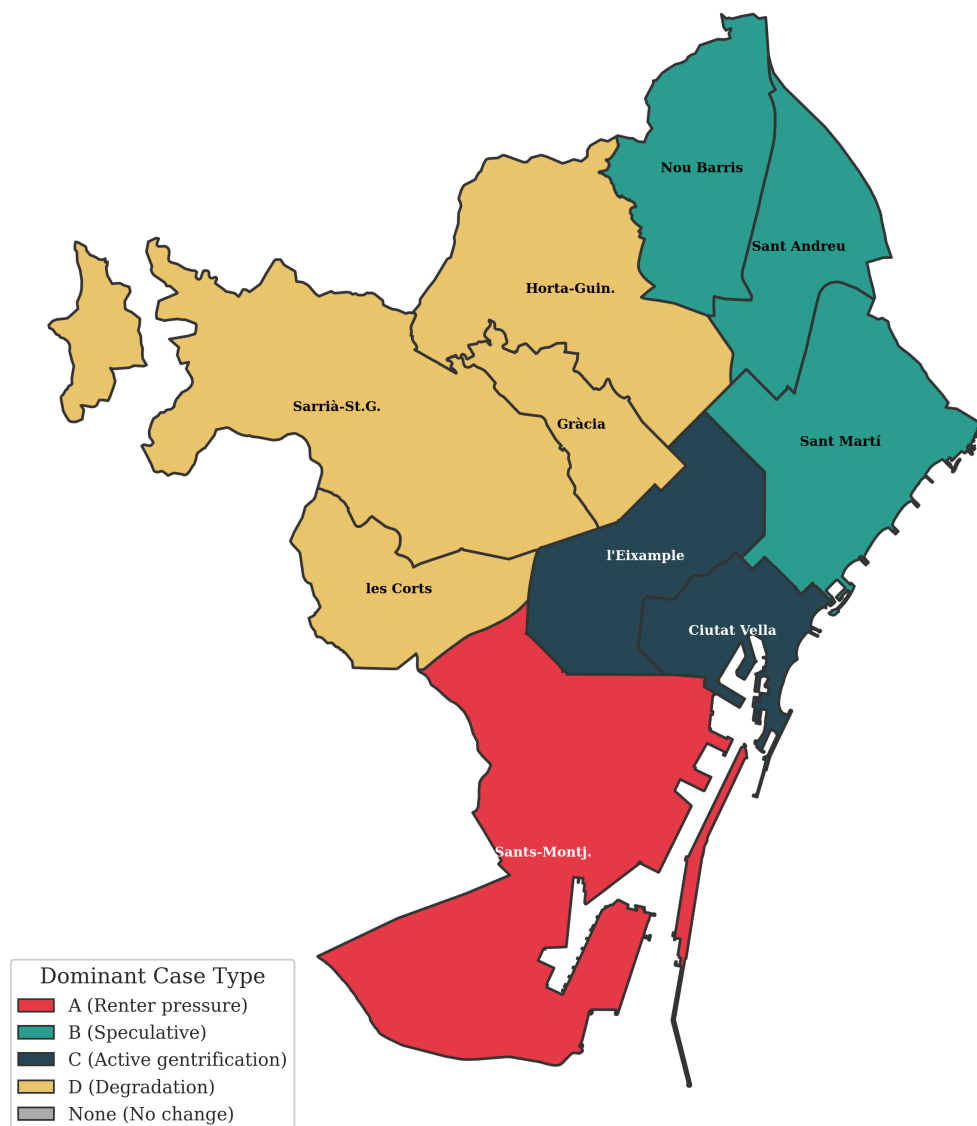


Figure 6: Dominant gentrification type by district in Barcelona (2012–2022). Case C concentrates in the historic center (Ciutat Vella, l'Eixample); Case B dominates working-class periphery (Nou Barris, Sant Andreu).

The temporal evolution at the district level (Figure 7) reveals a dramatic structural transformation. In 2012–2013, Case D dominated 20 of 21 Madrid districts and 9 of 10 Barcelona districts, reflecting the post-crisis collapse. By 2018–2019, Madrid had shifted almost entirely: Case C dominated 13 districts (all central) while Case B dominated 8 (all peripheral), with no district retaining D as its dominant type. Barcelona's transition was less complete: D remained dominant in 4 districts during 2018–2019, reflecting the city's stronger exposure to tourism-driven volatility. When the window is extended to include the COVID-19 disruption (2020–2021), Madrid reverts to B-dominant in 16 districts as the pandemic depressed rental markets, and Barcelona sees D re-expand to 8 of 10 districts.

Temporal Evolution of Dominant Gentrification Type by District

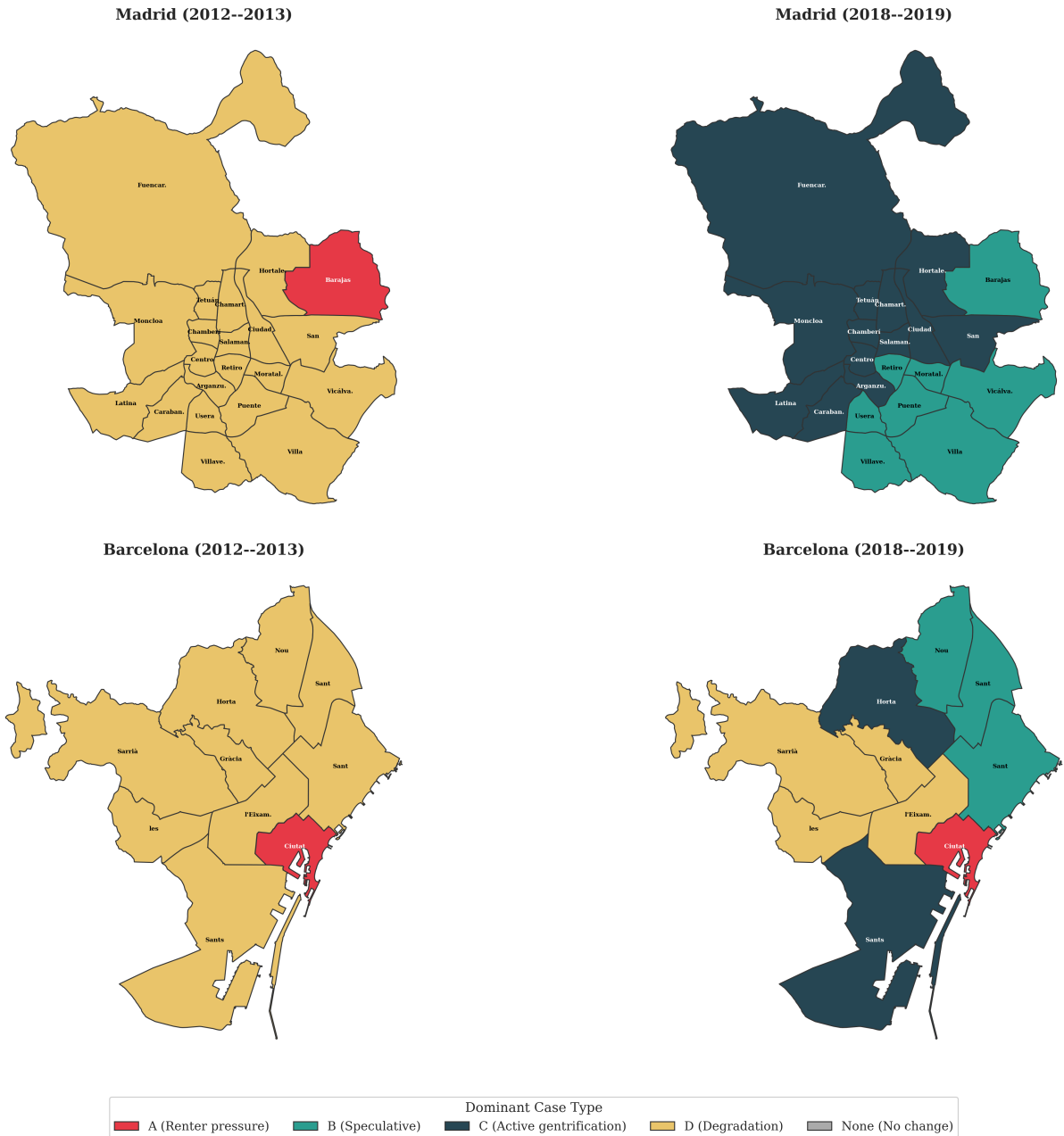


Figure 7: Temporal evolution of dominant gentrification type by district. In 2012–2013, degradation (Case D) dominated nearly all districts in both cities. By 2018–2019, Madrid had transitioned to a center–periphery pattern of active gentrification (C) versus speculation (B), while Barcelona retained more D-dominant districts.

This district-level center–periphery gradient is consistent with the classical gentrification sequence: central neighborhoods with cultural capital and infrastructure attract investment that drives simultaneous price increases (Case C), while peripheral areas experience the earlier stage of sale-driven speculation (Case B) that may or may not mature into full gentrification. The temporal shift from universal degradation (D) to differentiated patterns mirrors the rent-gap cycle at a finer geographic scale.

6 Discussion

The central finding is that gentrification trajectories are substantially predictable from housing market data. With rolling-origin retraining, regression meta-features, and directional decomposition, the model achieves 47.9% full-coverage accuracy, and selective prediction reveals a large subset where accuracy exceeds 58%. At 50% coverage the model achieves 58.6% on the five-class task; at 30% coverage, 64.7%. This reframes the practical value: rather than a census-wide oracle, the model functions as a triage tool that concentrates policy attention on the sections where prediction is reliable.

Rolling-origin retraining contributes the largest single gain (+2.3pp), demonstrating that concept drift across the COVID-19 disruption is a major source of error in fixed-split evaluation. By refitting the model each quarter using all available prior data, the classifier adapts to evolving market dynamics without any information leakage. The directional decomposition, which replaces the monolithic five-class classifier with two three-class price-direction models whose outputs are composed via the gentrification type definitions, adds a further 0.2pp by aligning the classification boundary with the underlying operationalization. Four additional robustness results reinforce the interpretation. First, the algorithm comparison demonstrates that the baseline ceiling is not an artifact of model choice. Second, external socioeconomic data adds negligible information. Third, the annual reformulation reproduces a comparable accuracy ceiling (42.4%). Fourth, spatial and history features add 1.5pp combined, confirming that neighborhood context carries genuine signal.

The dominance of current case type in feature importance reveals strong temporal persistence: gentrification, once initiated, tends to continue. This has direct policy implications: intervention is most effective before a neighborhood transitions from Case D (degradation) to Cases A or B. The temporal case distribution provides an empirical complement to Hackworth and Smith’s (2001) stage model: post-crisis degradation gives way to speculative investment as the rent gap widens (Smith, 2005), followed by consolidated gentrification. That LSTM networks underperform tree-based methods suggests that gentrification dynamics are better captured by feature-engineered snapshots than by raw sequential patterns. Limitations include the purely price-based operationalization, the 28% rental missing-data rate, and the imposition of discrete boundaries on continuous processes.

7 Conclusions

This study demonstrates that gentrification type is substantially predictable from census-section-level housing market data when framed as genuine one-step-ahead forecasting. An XGBoost pipeline with rolling-origin retraining and directional decomposition achieves 48% five-class accuracy ($2.4\times$ chance) on a temporally held-out test set spanning the COVID-19 disruption, and a binary displacement-risk detector achieves AUC 0.77 with 70% recall. Under selective prediction, accuracy reaches 58.6% at 50% coverage and 64.7% at 30% coverage, demonstrating that gentrification forecasting is practically reliable as a targeted policy tool. Future work should explore multi-quarter prediction horizons with smoothed targets and extend the analysis to other European housing markets.

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